



MECH MAINT Department
SIMHADRI



Ref: SMTTPP/MECH MAINT/2022-23/LMI/BMD/664351

Date: 09/12/2022(18:16:37)

Subject: Revised LMI on The Control of PFA Agglomerate Emissions

The LMI on "THE CONTROL OF PFA AGGLOMERATE EMISSIONS" LMI/OGN/OPS/MECH/011 has been revised as per revised OGN "COS-ISO-00-OGN/OPS/MECH/011" Rev. No.: 2 Oct. 2022. Revised LMI and OGN is attached.

Put up for kind approval please.

Submitted by:

कृष्ण लाल - वरिष्ठ प्रबंधक

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On: 09/12/2022(18:16:37) **Ref:** SMTTPP/MECH MAINT/2022-23/LMI/BMD/664351

MAY PLZ BE APPROVED.

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On:10/12/2022(11:26:25) **Ref:**SMTTPP/MECH MAINT/2022-23/LMI/BMD/664351

Following changes incorporated in revised LMI as per revised OGN:

1. Clause 7(III)-EPurging of Flakes by taking Fans in service removed.
2. Clause 7(III)-E revised to add Imported Coal and Duct cleaning.
3. Clause 7(III)-E- Water Jet washing in duct removed.

Revised LMI (attached doc "Revised LMI PFA Dec'22), attached as annexure. may please be approved.

Submitted by:

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Revised LMI may please be approved.

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On:17/12/2022(11:59:31) **Ref:**SMTPP/MECH MAINT/2022-23/LMI/BMD/664351

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/011/

ISSUE NO.2; REVISION NO. 5.00; DATE: DEC'2022

**THE CONTROL OF
PULVERISED FUEL ASH (PFA)
AGGLOMERATE EMISSIONS**

APPROVED FOR
Implementation

BUH, SIMHADRI

Date -----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	THE CONTROL OF PFA AGGLOMERATE EMISSIONS
<u>2.a) DOCUMENT CODE</u>	LMI/OGN/OPS/MECH/011 /
b) REVISION/ISSUE NO.&DATE	ISSUE-2; REV.-5.00; DEC'2022
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>OGN/OPS/MECH/011 Rev 02 Oct'2022</u>
<u>4.PREPARED BY</u>	SIGNATURE: Krishan Lal, Sr. Mgr (BMD)
<u>5. CHECKED BY</u>	SIGNATURE: Ritesh Kumar, Sr Mgr (BMD)
<u>6. RE-CHECKED BY</u>	SIGNATURE: Om Prakash, AGM (BMD)
<u>7. REVIEWED BY</u>	SIGNATURE: MATHEW EIPE KOVOOR, GM (Maint)
<u>8. REVIEWED BY</u>	SIGNATURE: Sanjay Kumar Sinha, GM (O&M)
<u>8. APPROVED BY</u>	SIGNATURE: Girish Chandra Choukse, HOP (SSTPP)
<u>9.DATE OF ISSUE</u>	
<u>10. REVISION INDEX</u>	ENCLOSED.
<u>11.DISTRIBUTION LIST</u>	ENCLOSED.

NTPC	LMI	SIMHADRI
TITLE: THE CONTROL OF PFA AGGLOMERATE EMISSIONS REV: 5.00 DATE: DEC'2022		
REF NO. LMI/OGN/OPS/MECH/011/		PAGE : 2 OF 7
<u>2.0 INTRODUCTION</u> The coal (indigenous/blended) used for Simhadri project has sulphur content varying upto 0.5% and combustion of this fuel produces sulphur-di-oxide and sulphur-tri-oxide vapours as part of flue gases. Wherever the local gas temperatures fall below the acid dew point in the flue gas duct work or chimney acid condensation occurs. The resultant wet acidic film provides sites for the capture of particulates present in the flue gas flow causing the formation of pulverized fuel ash (PFA) agglomerates. These agglomerates which become hard & fragile readily detach from the flue gas ducts with changes in gas velocities or with thermal cycles and are subsequently emitted from the chimney or deposited on the duct floor or chimney base. Agglomerate flake emissions which principally occur during unit start-up fall out in the nearby area, usually within one KM of the stack. Considering the layout of the Simhadri project with its vicinity to road this can be a source of nuisance and can cause strong objection from locals populace. The objective of this LMI is to instruct and monitor the measures to be taken for minimizing emission of PFA agglomerates arising from the operation of this plant <u>3.0 SCOPE</u> The scope of this LMI is to detail the methods that are to be adopted for the control of PFA agglomerate emissions. <u>4.0 REFERENCE DOCUMENT</u> Operation guidance note No. COS-ISO-00-OGN/OPS/MECH/011 Rev no.2 Oct'2022. <u>5.0 SUPERCEDED DOCUMENT:</u> Approved LMI ref: LMI/OGN/OPS/MECH/011/ Issue No.1: Revision No 4.00; Date : JULY,2022 <u>6.0 SPECIFIC INSTRUCTIONS</u> The following three pronged approach is to be followed to control PFA agglomerate emissions - Minimizing the period for which flue gas exit temperatures are below acid dew point - Reducing air ingress into flue gas thereby minimizing localized reduction in temperatures. - Cleaning of duct internals and chimney bottom of ash agglomerates.		

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<u>7.0 METHODOLOGY OF IMPLEMENTATION & RESPONSIBILITY CENTERS</u>		
<u>I. MINIMIZING THE PERIOD FOR WHICH FLUE GAS EXIT TEMPERATURES ARE BELOW ACID DEW POINT</u>		
A) Taking SCAPH and keeping SCAPH in service as long as it is necessary during cold start-ups & regulating steam flow to SCAPH so as to maintain average cold end temperatures after APH above 70° C. Exception to the same to be recorded Responsibility--- Head of Operation B) During normal operation flue gas dampers outlet to APH's shall be modulated to maintain average cold end temperatures at each APH above 70°C. Exception to the same to be recorded. Responsibility---Head of Operation C) Inspection of SCAPH for leakages and attending the same during overhaul/ Shutdowns. Compliance for the same to be recorded. Responsibility--- Head of MM-RM D) Mechanical cleaning of SCAPH heat transfer surfaces during overhaul. Compliance for the same to be recorded. Responsibility--- Head of MM-RM		
<u>II. REDUCING AIR INGRESS INTO FLUE GAS THEREBY MINIMIZING LOCALISED REDUCTION IN TEMPERATURES.</u>		
A) Pre-overhaul tests to be carried out to identify the extent of air-heater seal Leakages. Responsibility--- Head of EEMG B) Air preheater seal inspection for damage/ clearances and subsequent replacement if necessary and seal setting during unit overhauls. Inspection and work activities to be recorded. Responsibility--- Head of MM-RM C) Identification of air ingresses into flue gas duct from the manholes/tapping points/NMEJS in pre-overhaul survey and attending the same during overhaul. Inspection of flue gas duct internals from APH to chimney to the extent feasible for erosion and taking preventive patch welding during overhaul. Inspection & work activities to be recorded. Responsibility--- Head of MM-RM D) Post overhaul survey to monitor the effectiveness of seal setting. Responsibility--- Head of EEMG		

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III. CLEANING OF DUCT INTERNALS & CHIMNEY BOTTOM OF ASH AGGLOMERATE

A) Cleaning of duct from APH to ESP of accumulated ash during overhaul. Compliance for the same to be recorded.
Responsibility--- Head of MM-RM

B) Cleaning of duct walls from ESP to chimney to the extent feasible & chimney bottom Agglomerates during overhauls. Compliance for the same to be recorded. -

Responsibility--- Head of MM-RM

C) Evacuation of Eco-hoppers on regular basis and to be monitored on shift wise.

Responsibility--- Head of AHP/Operation

D) Dosing of Ammonia in Flue gas based on emission levels (>80mg/nm³)

Responsibility - Head of OPERATION

E) If evidence of significant agglomerate emissions obtained or during firing of high Sulphur Coals like in imported coal, advantage of plant shutdown to be taken to clean ductwork walls by manual brushing and to remove flakes and debris from the bottom of ductwork prior to the re-start of the unit fans.

Whilst agglomerate flaking most often occurs when plant is taken off load, hence if prior history of flaking is observed, these flakes and debris are to be removed from bottom of duct before restarting.

Responsibility--- Head of MM-RM

8.0 EXCEPTION REPORTING

1. The following exception reporting shall be done. Responsibility--- Head of Operation

Exception	Unit	Date	Remark
Observation of PFA agglomerate emissions			
SCAPH not in service during cold start-up			
Average cold end temperatures less than 70 deg during normal			

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REV.5.00 DEC'22				
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REVISION INDEX (for changes/revision)				
SL No.	Rev. no.	Date	Details	Approved by
1.	REV 1.00	JUL'06	Coal changes indicated	
			Reviewing authority revised	
			Distribution list revised	
			Repair action indicated in pt 6 of compliance record	
2.	REV 2.00	JUL'13	Main page (Rev & BUH)	
			LMI approval documents page (S.No.2b,4,5,6)	
			Page 2 (Sec 5)	
			Page 4 (Sec 11)	
3.	REV 3.00	DEC'17	LMI approval documents - S.No. 2b rev no change, 5 added, 6-Desg. Change)); Page-2, cl.5 rev no. change; Page 6 (2,3,4,6,8,9- design. change, 5 & 10 added)	
4.	Rev 4.00	July'20	III .Cleaning of duct Section 7. III E Section 7. III F Page No 4 (sec 7.III E) Page No 4 (sec 7.III F)	
5.	Rev 4.00	July'22	Point II. C correction of NMEPS to NMEJs	
6.	Rev 5.00	Dec'22	Clause 7.III.E- Revised and purging of flakes by taking Fan in Service removed. Clause 7.III.E- Revised to add imported coal and duct cleaning Clause 7.III.F- Water Jet washing in duct removed	

[illegible]

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TITLE: THE CONTROL OF PFA AGGLOMERATE EMISSIONS		
REV: 5.00 DATE: DEC'2022		
REF NO. LMI/OGN/OPS/MECH/011/		PAGE : 7 OF 7
DESCRIPTION		COMPLIANCE
1. PRE-OVERHAUL SURVEY TO IDENTIFY AIR INGRESS INTO FLUE GAS DUCT SYSTEM CARRIED OUT 2. AIR-PREHEATER SEAL INSPECTION & REPLACEMENT/SETTING CARRIED OUT DURING OVERHAUL. 3. SCAPH INTERNAL INSPECTION FOR LEAKAGES & SUBSEQUENT RECTIFICATION AND CLEANING OF HEATEXCHANGER SURFACES CARRIED OUT DURING OVERHAUL 4. CLEANING OF DUCT FROM APH TO ESP OF ACCUMULATED ASH CARRIED OUT DURING OVERHAUL. 5. CLEANING OF DUCT WALLS FROM ESP TO CHIMNEY OF AGGLOMERATES CARRIED OUT DURING OVERHAUL 6.FLUE GAS DUCT FROM APH TO CHIMNEY, INTERNAL INSPECTION CARRIED OUT FOR DUCT EROSION DURING OVERHAULREPAIR OF IDENTIFIED LOCATIONS.		

